

Miaomiao Dai

Ann Arbor, MI · dmsjtum@umich.edu · Personal website: dmsjtum.github.io

EDUCATION

University of Pennsylvania

M.S.E. in Robotics

Aug 2026 – present

Philadelphia, PA

University of Michigan

B.S.E. in Computer Science, Summa Cum Laude (SJTU–UMich dual-degree sequence) · GPA 3.8/4.0

Aug 2024 – May 2026

Ann Arbor, MI

- Coursework: Robot Learning (A+), Algorithmic Robotics (A), Machine Learning (A), Computer Vision (A), Data Structures & Algorithms (A)

Shanghai Jiao Tong University

B.E. in Mechanical Engineering, Global College · GPA 3.6/4.0

Aug 2022 – Aug 2024, May 2026 – Aug 2026

Shanghai, China

PUBLICATIONS

- [1] **Miaomiao Dai**, Zhongqiang Ren. “HOP: Fast Differential Dynamic Programming for Horizon-Optimal Trajectory Planning.” *Robotics: Science and Systems (RSS)*, 2026.
- [2] **Miaomiao Dai**, Zhongqiang Ren. “AL-HOP: Augmented-Lagrangian Horizon-Optimal Planning for Fast Constrained Trajectory Optimization.” In submission to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027.

RESEARCH EXPERIENCE

HOP: Horizon-Optimal Trajectory Planning

Jul 2025 – Feb 2026

Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · RSS 2026 talk, undergraduate first author

- Standard trajectory optimizers fix the task duration in advance and re-solve for every candidate; HOP selects the optimal horizon automatically within a single solve.
- Proposed and proved a **closed-form theory** that scores every candidate horizon in one backward sweep, cutting horizon search from $O(N^2)$ to $O(N)$ — up to **40×** faster than brute force, with identical selected horizon and cost.
- Designed and iterated the experimental evaluation across cartpole, Segway, and quadrotor systems.

AL-HOP: Constrained Horizon-Optimal Planning

Mar 2026 – present

Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · first-author manuscript in submission to ICRA 2027

- Extends HOP to hard state–control constraints (obstacles, actuator limits) with an augmented-Lagrangian formulation that keeps the one-sweep horizon prediction.
- Produced solutions valid under the full nonlinear dynamics on **100/100** benchmark problems across 10 dynamical systems, vs. 79, 71, and 55 for ALTRO and direct collocation (IPOPT, SNOPT); where both were valid, baselines needed $1.6\text{--}3.1\times$ the solve time.

Touch Without Touch: Measured-Touch Supervision for Visual Representations

Jun 2026 – present

Advisors: Prof. Hesheng Wang, IRMV Lab, Shanghai Jiao Tong University · Prof. Lin Shao, RoboScience

- Asks whether paired, *measured* touch — pressure, force, contact — can teach an RGB representation physical-interaction structure that visual self-supervision alone does not; inference stays strictly **RGB-only** at deployment.
- **Preregistered the protocol** before training: controlled comparisons isolating pairing correctness, sensor fidelity, and how touch enters the model, each with falsification criteria frozen in advance and evaluated by zero-shot human-to-robot transfer.
- **Built the training stack**: JEPA-style latent prediction with an EMA teacher over frozen visual and tactile features; unified **five** heterogeneous touch datasets (human hand pressure, tactile arrays, GelSight imagery, force–torque) into one pipeline running across a local RTX 5090 and a multi-GPU A100 server.

STA: Physical Supervision from Human Video for Vision–Language–Action Policies

Aug 2026 – present

Advisors: Prof. Hesheng Wang, IRMV Lab, Shanghai Jiao Tong University · Prof. Lin Shao, RoboScience

- Tests whether the physical structure carried by human video — contact, and how a motion loads over time — can supervise a vision–language–action policy, with **no extra sensor at deployment**.
- Kept the physical pathway **strictly decoupled**: it reads only the current image and the raw instruction through a frozen visual encoder, never the policy’s own hidden states, and enters the action path as one gated residual that the robot-only stage freezes — so a gain cannot be the policy re-reading what it already knew.
- The two arms run identical modules and differ in **one loss term**. At the current scale the physical term is a **clear causal gain** over its isomorphic control, and shuffling the target in time or pairing it with the wrong window removes most of the effect; the decisive larger-scale run is pending.

- **Collected and trained on real robot data:** teleoperated **316 episodes** across 11 tabletop tasks on a low-cost SO-101, then fine-tuned $\pi_{0.5}$ on them — LoRA on the 2B vision–language backbone, full training of the 300M action expert.
- **Closed the execution loop:** a VLA emits open-loop action chunks and never checks the result, so I added a lower-rate visual supervisor over progress and terminal state that triggers stop, return, re-observation, or a bounded retry (terminal-verification F_1 **0.842**).
- Reached **70%** end-to-end success over 11 tasks (85% pick, 75% placement) with zero safety incidents; unseen object–target compositions are reported separately as an open condition, and built the voice-interaction and monitoring interface.

HONORS & AWARDS

James B. Angell Scholar, University of Michigan	Mar 2026
University Honors, University of Michigan (3 terms)	Dec 2024 – Dec 2025
Dean’s List, University of Michigan (3 terms)	Dec 2024 – Dec 2025
Honorable Mention, Interdisciplinary Contest in Modeling (ICM)	May 2023
First Prize, 13th SJTU Liming Cup Mechanical Innovation Competition	May 2023
Outstanding Volunteer, Miyuan Volunteer Team	Mar 2023

SKILLS

Methods: trajectory optimization (DDP, augmented-Lagrangian methods), self-supervised representation learning (JEPA-style latent prediction), VLA fine-tuning & real-robot deployment ($\pi_{0.5}$), VLM integration

Languages: Python, C++, C, MATLAB

ML & robotics: PyTorch, Transformers, OpenCV, NumPy, SciPy, OpenAI Gym, ROS, PyBullet, Gazebo

Tools: Git, Linux (Ubuntu), Docker, $\LaTeX$, SolidWorks